Detailed Operating System Notes

Prepared for Project PDF Testing and Academic Study

1. Introduction to Operating Systems

An Operating System (OS) is system software that acts as an interface between users and computer hardware. It manages resources such as CPU, memory, storage, and I/O devices. Major functions include process management, memory management, file management, device management, security, and user interaction. Examples include Windows, Linux, macOS, Android, and iOS. Objectives of an OS are convenience, efficiency, security, and reliability.

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2. Types of Operating Systems

Batch OS executes jobs in batches without user interaction. Multiprogramming OS keeps multiple jobs in memory. Multitasking OS allows users to run multiple applications simultaneously. Time-sharing OS allocates CPU time slices among users. Real-Time OS guarantees response within deadlines. Distributed OS coordinates multiple computers as one system.

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3. Process Management

A process is a program in execution. Process states include New, Ready, Running, Waiting, and Terminated. The Process Control Block (PCB) stores process information such as process ID, state, registers, and memory details. Context switching occurs when the CPU switches from one process to another.

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4. CPU Scheduling

CPU scheduling determines which process gets CPU time. Goals include maximizing CPU utilization and minimizing waiting time. FCFS executes processes in arrival order. SJF selects the shortest job. Round Robin assigns fixed time quanta. Priority Scheduling executes the highest-priority process first.

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5. Threads and Multithreading

A thread is the smallest unit of CPU execution. Multiple threads within a process share resources. Benefits include improved responsiveness, resource sharing, and better performance on multicore systems.

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6. Memory Management

Memory management allocates and deallocates memory to processes. Fixed partitioning divides memory into fixed blocks, while dynamic partitioning allocates memory according to process needs. Fragmentation can be internal or external.

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7. Virtual Memory and Paging

Virtual memory allows execution of large programs using secondary storage as an extension of RAM. Paging divides logical memory into pages and physical memory into frames. It reduces external fragmentation and improves memory utilization.

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8. File Systems

A file system organizes and manages data stored on disks. Common operations include create, read, write, update, rename, and delete. File access methods include sequential, direct, and indexed access.

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9. Deadlocks

A deadlock occurs when processes wait indefinitely for resources held by one another. Four necessary conditions are mutual exclusion, hold and wait, no preemption, and circular wait. Deadlocks can be prevented, avoided, detected, or recovered from.

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10. Security and Protection

Security protects systems from unauthorized access. Mechanisms include authentication, authorization, encryption, firewalls, and access control. Protection controls how processes access system resources.

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Important Exam Questions

1. Define Operating System and explain its functions.
2. Explain different types of Operating Systems.
3. What is a Process? Explain PCB.
4. Compare FCFS, SJF, and Round Robin scheduling.
5. Explain Memory Management and Fragmentation.
6. What is Virtual Memory? Explain Paging.
7. Define Deadlock and explain its conditions.
8. Explain File Systems and File Access Methods.
9. Discuss Security and Protection in OS.
10. Explain Real-Time Operating Systems.